Int J Clin Pharm
DOI 10.1007/s11096-015-0210-4

RESEARCH ARTICLE

Pharmaceutical care of elderly patients with poorly controlled type 2 diabetes mellitus: a randomized controlled trial

Jyun-Hong Chen[1,2,3] · Huang-Tz Ou[3] · Tzu-Chieh Lin[3] · Edward Chia-Cheng Lai[3] · Yea-Huei Yang Kao[3]

Received: 22 April 2015 / Accepted: 7 October 2015
© Koninklijke Nederlandse Maatschappij ter bevordering der Pharmacie 2015

Abstract *Background* Care of the elderly with diabetes is more complicated than that for other age groups. The elderly and/or those with multiple comorbidities are often excluded from randomized controlled trials of treatments for diabetes. The heterogeneity of health status of the elderly also increases the difficulty in diabetes care; therefore, diabetes care for the elderly should be individualized. Motivated patients educated about diabetes benefit the most from collaborating with a multidisciplinary patient-care team. A pharmacist is an important team member by serving as an educator, coach, healthcare manager, and pharmaceutical care provider. *Objective* To evaluate the effects of pharmaceutical care on glycemic control of ambulatory elderly patients with type 2 diabetes. *Setting* A 421-bed district hospital in Nantou City, Taiwan. *Method* We conducted a randomized controlled clinical trial involving 100 patients with type 2 diabetes with poor glycemic control (HbA1c levels of $\geq$9.0 %) aged $\geq$65 years over 6 months. Participants were randomly assigned to a standard-care (control, n = 50) or pharmaceutical-care (intervention, n = 50) group. Pharmaceutical care was provided by a certified diabetes-educator pharmacist who identified and resolved drug-related problems and established a procedure for consultations pertaining to medication. The Mann–Whitney test was used to evaluate nonparametric quantitative data. Statistical significance was defined as $P < 0.05$. *Main outcome measure* The change in the mean HbA1c level from the baseline to the next level within 6 months after recruiting. *Results* Nonparametric data (Mann–Whitney test) showed that the mean HbA1c level significantly decreased (0.83 %) after 6 months for the intervention group compared with an increase of 0.43 % for the control group ($P \leq 0.001$). Medical expenses between groups did not significantly differ (−624.06 vs. −418.7, $P = 0.767$). There was no significant difference in hospitalization rates between groups. *Conclusion* The pharmacist intervention program provided pharmaceutical services that improved long-term, safe control of blood sugar levels for ambulatory elderly patients with diabetes and did not increase medical expenses.

Keywords Clinical trials · Diabetes · Elderly · Pharmaceutical care · Taiwan

Impact of findings on practice

- Including well-trained pharmacists as members of a multidisciplinary diabetes care team (including physicians, certified diabetes-educator nurses, and dietitians) improved long-term, safe control of blood sugar levels of poorly controlled ambulatory elderly patients with diabetes.
- Including a pharmacist case manager in the diabetes care team of a Pay-for-Performance program could improve the quality of care and provide a beneficial long-term outcome without increasing expenses.

✉ Yea-Huei Yang Kao
yhkao@mail.ncku.edu.tw

[1] Department of Pharmacy, Nantou Hospital, Ministry of Health and Welfare, Nantou City, Taiwan, ROC

[2] Isotope Application Division, Institute of Nuclear Energy Research, Atomic Energy Council, Executive Yuan, Taipei City, Taiwan, ROC

[3] Institute of Clinical Pharmacy and Pharmaceutical Sciences, College of Medicine, National Cheng Kung University, No. 1, University Road, Tainan City 701, Taiwan, ROC

Published online: 24 October 2015

Introduction

Diabetes mellitus (DM) is a chronic illness requiring continuous medical care and ongoing patient self-management education and support to prevent acute complications and reduce long-term complication risk [1]. Worldwide, the number of people with DM was projected to increase from 171 million in 2000 to 366 million in 2030 [2]; WHO projects that diabetes will be the seventh leading cause of death in 2030 [3]. The most important demographic change in global DM prevalence is the increased vulnerability of the proportion of people aged >65 years to the disease, e.g., 26.9 % of all people aged >65 years in the United States had diagnosed and undiagnosed DM in 2010 [4]; DM prevalence was >20 % in patients aged ≥60 years in Taiwan in 2004 [5]. There were >85 % patients aged >60 years among diabetes population; there were >15,000 new cases in Taiwan in 2008 [6]. These numbers can be expected to rapidly increase in the coming decades.

DM is an important health condition for the aging population because it increases premature death risk, functional disability, and comorbidities, including hypertension and chronic heart disease [7]. Older adults (≥65 years) with DM are at greater risk than other older adults for polypharmacy and several common age-related conditions, including depression, cognitive impairment, urinary incontinence, injury from falls, and persistent pain.

The American Diabetes Association convened a Consensus Development Conference on diabetes and older adults (≥65 years) in February 2012 [8] that produced a framework for considering treatment goals for hyperglycemia, high blood pressure, and dyslipidemia and provided consensus recommendations for care. They recommend that DM self-management education/training (DSME/T) for older adults should be individualized. In particular, DSME/T should consider possible sensation impairments, vision, hearing, cognition, and functional/physical status. Further recommendations include aligning DM care with components of the Chronic Care Model to achieve successful outcomes when a prepared and proactive team interacts with an informed and engaged patient. Incorporating physicians, nurses, pharmacists [9], and other providers in care-management teams optimizes the performances of the healthcare provider and team, leading to reduction in HbA1c levels, blood pressure, and low-density lipoprotein cholesterol [10, 11].

Pharmaceutical care for diabetic patients improved clinical values, reduced HbA1c levels, improved self-management, increased DM knowledge [12, 13], improved blood pressure management [14], and reduced cardiovascular-risk scores [15]. Prospective studies determined patients' HbA1c values before and after a pharmacist were consulted [16, 17]. Randomized controlled trials that compared DM care provided by pharmacists with that provided by diabetes care team members, including physicians, nurses, or dietitians, did not specifically consider elderly patients [10, 18]. Otherwise, the medical expense of pharmacist involved in the team has not been previously discussed.

Aim of the study

This study aimed to assess changes in HbA1c levels and medical expenses of patients managed by a diabetes care team that included a pharmacist compared with those of patients who received standard care (without a pharmacist).

Ethical approval

The study was approved by the institutional review board of the Taoyuan General Hospital, Ministry of Health and Welfare, Taiwan, ROC and registered at clinicaltrials.gov (NCT01455727).

Methods

Study design and setting

We conducted a randomized controlled trial from August 2011 to February 2012. The primary outcome was HbA1c level change within 6 months. We administered the Mini-Mental State Examination (MMSE) [19] to assess cognition and the Taiwan Geriatric Depression Scale-Short Form (TGDS-S) [20] to screen for depression.

The study was conducted at the Nantou Hospital, Ministry of Health and Welfare, Taiwan, ROC, which participates in the Pay-for-Performance (P4P) Program for DM care in Taiwan. DM is among five diseases included in the P4P program designed by the National Health Insurance (NHI) in Taiwan since 2001 [21]. The NHI's P4P program for DM care provides financial incentives for healthcare providers to increase comprehensive follow-up visits, including enhanced self-care education and annual diabetes-specific physical examinations, including eye examinations and laboratory tests (e.g., HbA1c test). The diabetes care team of the P4P Program includes physicians, certified diabetes-educator (CDE) nurses, and dietitians.

Nantou Hospital's patients participating in the P4P program were enrolled from thirteen villages and towns in the Nantou City area; 71.9 % of patients in the Nantou Hospital were enrolled in 2010. The data were recorded in the computerized physician order entry system and the

Taiwan Diabetes Shared Care Program (TDSCP) database [22].

Patients and randomization

Ambulatory patients with DM who were aged ≥65 years with HbA1c levels ≥9.0 % (75 mmol/mol) in 2010 were referred to the diabetes care pharmacist by the CDE nurse who was the P4P program's case manager. The pharmacist interviewed patients in an independent office, and informed consent, which was required by the institutional ethics review board, was obtained from patients who agreed to participate in the study. Patients were excluded if they were diagnosed with cancer or if they refused to participate after the pharmacist explained the details and questions regarding the study.

Using random numbers generated by SAS 9.2 (for Windows), the case manager assigned patients to the intervention or control group before the nurse referred patients to the pharmacist. The severity of patient comorbidity was calculated using the Charlson comorbidity index. Comorbidity data were obtained from the Nantou Hospital's NHI reimbursement data and the computerized physician order entry system.

Intervention

A pharmacist provided all pharmaceutical and diabetes-related care for the intervention group. The pharmacist was also a DM educator certified by the Taiwanese Association of Diabetes Educators. According to the "Standards of Medical Care in Diabetes—2014 and Guidelines for Improving the Care of the Older Person with Diabetes Mellitus" guidelines [7], the care of older adults with DM can be complicated because of their clinical and functional heterogeneity. If there is an evidence of difficulty with adherence to a regimen that cannot or should not be simplified, a physician, pharmacist, DM educator, or other healthcare practitioner should provide counseling to the patients and/or family members. Therefore, the pharmaceutical care provided to the intervention group by the pharmacist comprised the assessment of adherence to pillbox use and insulin injection technique and the appropriateness of their current medication regimens and follow-up. Evaluating cognition and screening for depression using the MMSE and TGDS-S, respectively, would help the pharmacist identify and resolve drug-related problems. Diabetes-related care was provided by the pharmacist as well, including appropriate individual DM education, recommendations to the patients' physicians, and referral of patients to other diabetes care team members, including the CDE nurse and dietitians. Changes to medication regimens were confirmed by their prescribing physician after counseling by the pharmacist.

Follow-up visits were supplemented with monthly telephone calls, as required for medication counseling and adverse event surveys. Patients were personally visited, if required or if they made an appointment when they returned to the hospital to renew their monthly prescription. Face-to-face visits were conducted at least once at enrollment and 1.86 times on an average during the study period; after enrollment, monthly telephone calls were conducted 2.48 times on an average. In the P4P program, patients' HbA1c levels were measured every 3 or 6 months during follow-up. After enrollment, patients were followed up for 6 months; then, their second HbA1c assay was performed. Laboratory data were collected from the computerized physician order entry system. Insurance claim data were obtained with the permission of the Director, Nantou Hospital.

Study outcomes

The primary outcome was defined as the mean change in HbA1c levels from baseline to the next level within 6 months after the study commenced. HbA1c level represents an important marker of disease control that is negatively associated with long-term diabetes-related morbidity. Glycemic control goals for certain subjects may be reasonably relaxed, according to individualized criteria. Older adults who are functionally and cognitively intact with significant life expectancy should receive DM care using goals developed for younger adults [6]. According to the framework for considering treatment goals for hyperglycemia in older adults with DM, the HbA1c target-level should be <8.5 %.

The secondary outcome variable, financial analysis, was the difference of total insurance payment points for ambulatory care services, pharmacist intervention cost, and hospitalization rates before and after the intervention. During 2011 to 2012, the ratio of the New Taiwan Dollar (NTD) to payment points was approximately 0.92, which fluctuated every 3 months. For evaluating patients' medical expenses, the payment points claimed for health insurance were a good surrogate of patients' medical expenses. The data was obtained from the hospital information system (HIS) database. Pharmacist intervention cost was estimated as the pharmacist's salary, telephone fees, and the cost of supplies (e.g., paper, education handouts, pillboxes). The cost-effectiveness and safety of pharmacist intervention was also analyzed.

Statistical analysis

A power analysis using G*Power 3.0.3 [23] indicated that ≥37 patients were required in each study group to detect a clinically significant decrease of 1.0 % ($1 - \beta = 0.80$ and $\alpha = 0.05$) in HbA1c level. The χ^2 (Chi-square) and Fisher's exact test were considered appropriate for nominal data. The

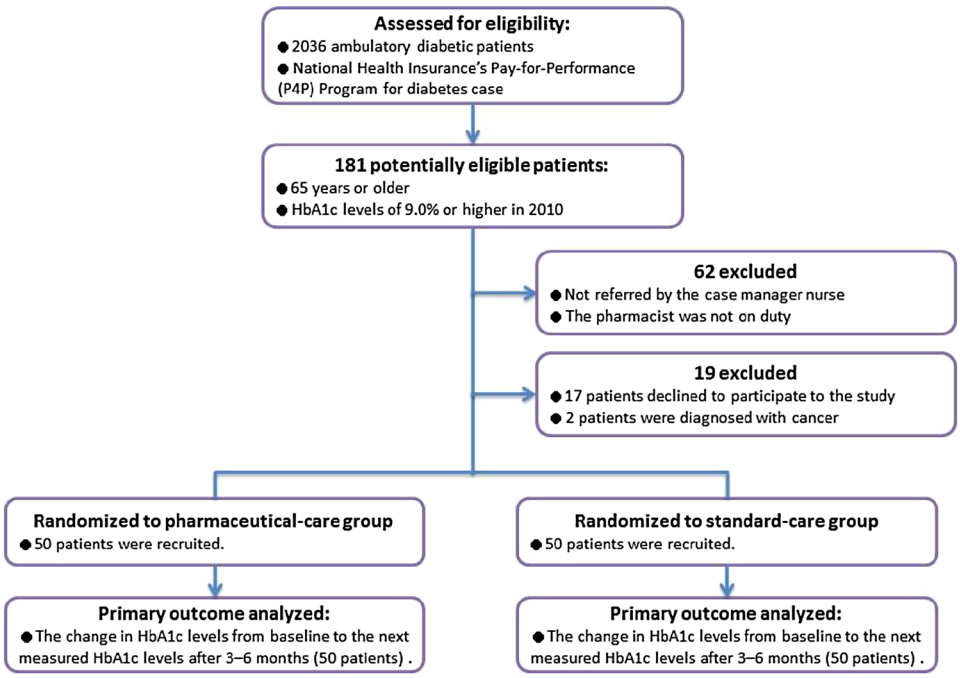

Fig. 1 Study flowchart

Mann–Whitney test was used to evaluate nonparametric quantitative data. Statistical significance was defined as $P < 0.05$. The statistical software used was SPSS for Windows, version 17 (SPSS Inc., Chicago, IL, USA).

Results

Data collected from January 2009 from the TDSCP database and Nantou Hospital's electronic medical records identified 2036 ambulatory patients with DM. The recruitment began in August 2011; the final follow-up of the last patient was performed in February 2012. There were 181 potentially eligible patients aged ≥65 years who had HbA1c levels ≥9.0 % (75 mmol/mol) in 2010 who were referred to the ambulatory case-manager nurse and were randomly assigned to either group. Of these, 119 patients were referred to the pharmacist and invited to participate. Sixty-two patients who were not referred by the case manager nurse or the pharmacists who were not on duty at that time were excluded. We also excluded 17 patients who declined to participate and 2 patients who were diagnosed with cancer. One hundred patients completed the study (control group, 50/61; intervention group, 50/58) (Fig. 1).

The flow of participants through the study is shown in Fig. 1. The demographic and clinical characteristics of the groups were very similar (Table 1). The severity of patients' comorbidities was not significantly different between groups (3.22 and 3.62, respectively, $P = 0.228$). In contrast, the intervention group patients' body weights were significantly higher ($P = 0.02$). The intervention group's HbA1c level was higher (9.22 %, 77 mmol/mol vs. 8.94 %, 74 mmol/mol), although the difference was not statistically significant ($P = 0.408$).

We obtained baseline and final HbA1c values 6 months after recruiting all patients. There was no difference between groups in fasting plasma glucose levels at the beginning of the study. Patients' HbA1c levels in the intervention group significantly decreased during 6 months (−0.83 %, $P < 0.001$). In contrast, the control group's HbA1c level significantly increased (0.43 %, $P < 0.001$) (Table 2). With regard to individual patient's improvement, a statistically significant higher percentage of patients in the intervention group had reduced HbA1c levels ≥0.5 % (54 %, intervention group; 8 %, control group).

Subgroup analysis revealed that the men and women patients' baseline HbA1c values in the intervention group were not significantly different from those in the control group [men, 9.10 % (76 mmol/mol) vs. 8.98 % (75 mmol/mol), $P = 0.884$; women: 9.35 % (79 mmol/mol) vs. 8.90 % (74 mmol/mol), $P = 0.356$, for intervention and control groups, respectively]. Men and women patients' HbA1c levels in the intervention group significantly decreased (men, −0.64 %, $P < 0.001$; women, −1.03 %, $P < 0.001$).

The intervention group's medical expenses were not significantly different from those of the control group (−624.06 vs. −418.75, $P = 0.767$) (Table 2). The hospitalization rate for each group decreased but not significantly (−0.14 vs. −0.02 %, $P = 0.654$). Hypoglycemia is an important adverse effect of glucose-lowering agents. Hospitalizations due to hypoglycemia were observed for

Table 1 Baseline characteristics of patients

	Intervention group (n = 50)	Control group (n = 50)	*P* value
1. Mean age	72.16 (6.6)	72.76 (5.9)	0.635
2. Gender (male:female) (%)	(50:50)	(50:50)	
3. Mean weight (kg)	66.62 (12.0)	61.02 (11.7)	0.02*
4. BMI	26.58 (4.2)	25.00 (3.5)	0.059
5. Systolic blood pressure (mmHg)	135.08 (18.2)	132.14 (16.7)	0.535
6. Diastolic blood pressure (mmHg)	74.74 (10.8)	75.12 (10.8)	0.629
7. Mean fasting plasma glucose (mg/dl)	204.62 (91.3)	190.08 (85.4)	0.35
8. Education			0.642
No high school degree (%)	39 (78)	41 (82)	0.617
High school degree (%)	8 (16)	5 (10)	0.372
Academic degree (%)	3 (6)	4 (8)	1.0
9. Smokers (%)	7 (14)	4 (8)	0.338
10. Drinkers (%)	3 (6)	6 (12)	0.487
11. Oral antidiabetic medication (%)			
Metformin	35 (70)	33 (66)	0.668
Sulphonylureas	37 (74)	35 (70)	0.656
Glinides	3 (6)	2 (4)	1.0
α-Glucosidase inhibitor	11 (22)	8 (16)	0.444
TZDs	21 (42)	19 (38)	0.683
DPP-4 inhibitor	2 (4)	0 (0)	0.495
12. Oral antidiabetic agents per patient	2.18	1.94	0.281
13. Insulin users (%)	19 (38)	19 (38)	1.0
14. Lipid-lowering agent users (%)	22 (44)	21 (42)	1.0
15. Antihypertensive agent users (%)	36 (72)	36 (72)	1.0
16. Mean duration of DM	13.36 (6.9)	14.84 (5.6)	0.218
17. HbA1c (mmol/mol) (SD)	9.22 (77) (1.7)	8.94 (74) (1.5)	0.408
18. MMSE score (SD) (%)	25.43 (4.1) (60)	25.29 (3.2) (54)	0.52
19. GDS score (SD) (%)	2.94 (3.0) (68)	3.07 (3.4) (60)	0.93
20. Charlson comorbidity index	3.22 (1.58)	3.62 (1.69)	0.228

Data are presented as the mean (SD) or number of participants (%)

Table 2 Comparison of HbA1c levels and payment points before and after intervention

Variable	Intervention group (50)	Control group (50)	*P* value*
Mean HbA1c % (mmol/mol) (SD)			
Before intervention	9.22 (77) (1.7)	8.94 (74) (1.5)	0.408
After intervention	8.39 (68) (1.2)	9.37 (79) (1.5)	0.002*
Difference between after and before intervention[a]			
HbA1c % (SD)	−0.83 (1.2)	+0.43 (0.87)	<0.001*
Mean payment points (SD)			
Before intervention	11,830.54 (8493.0)	11,371.86 (11,387.1)	0.825
After intervention	11,206.48 (7581.5)	10,953.16 (9154.4)	0.863
Difference between after and before intervention[a]			
Payment points (SD)	−624.06 (5184.6)	−418.75 (5074.3)	0.767

Data are presented as the mean (SD)

[a] (−) decrease in value after intervention, (+) increase in value after intervention

* Mann–Whitney test

Table 3 Costs of pharmaceutical care

Cost parameter	Cost per unit	Mean cost per intervention group patient	Mean cost per control group patient
Pay for pharmacist	244 NTD/h	1220 NTD/5 h (initial and follow-up visits for questionnaire, conducting medication reviews, drug utilization evaluation, consultation with physician or dietitian, patient education, and telephone contact)	122 NTD/30 min (initial and follow-up visits for questionnaire)
Telephone fee	Cellphone: 5.16 NTD/min Fixed-line telephone: 0.53 NTD/min	56.9 NTD/20 min (approximately half of the patients were contacted by cell phone, others by fixed-line telephone)	
Pillbox	30 NTD/box	30 NTD/box (provided to patients of the intervention group, 2 types of pillboxes: 8 blocks with fixed size or 6 blocks with different size)	
Papers and educational handouts	2 NTD/page	30 NTD/15 page (including the copy of personal data sheet, questionnaire, and printout of education handouts)	10 NTD/5 page (including the copy of personal data sheet, questionnaire)
Total cost		1336.9 NTD/patient	132 NTD/patient

6 months in both groups before and after recruitment. After recruitment, one patient in the control group required hospitalization due to hypoglycemia.

From the third-party payer perspective, we estimated the expenditure for pharmacist intervention for both groups (Table 3). All expenditures in the study project were funded by the Department of Health, Executive Yuan, ROC. The mean reduction in HbA1c levels achieved by pharmaceutical care compared with that of standard care was approximately 0.83 %. The mean cost per patient in the intervention group was NTD 1336.9 versus NTD 132 in the control group. This represented an increase in cost of NTD 1204.9 per patient. The net cost and the net effectiveness of the intervention group compared with the control group were calculated and expressed as a cost-effectiveness ratio (the additional cost per benefit gained). Thus, the incremental cost per 1 % reduction in the mean HbA1c level for pharmaceutical care compared with standard care was NTD 1451.69. From the hospital manager's perspective, if patients were satisfied with their health outcomes (decrease in HbA1c level or absence of adverse drug events) achieved by the pharmacist-involved diabetes care team, they may increase their follow-up visits. Because the P4P program provides financial incentives for hospitals for patients' complete follow-up, the hospital would receive more payment points from health insurance reimbursement, decreasing the additional pharmaceutical care service costs.

Discussion

We observed a significant reduction in the mean HbA1c levels from 9.22 % (77 mmol/mol) to 8.39 % (68 mmol/mol) in the pharmaceutical care group. The United Kingdom Prospective Diabetes Study suggests that a 1 % reduction in HbA1c level reduces macrovascular complications, including myocardial infarction, stroke, and amputation (peripheral vascular disease) by 21 % [24]. However, the most appropriate range of glycemic levels associated with the lowest complications and mortality rates in older patients with DM is controversial.

A retrospective cohort study using Kaiser Permanente Northern California Diabetes Registry [25] enrolled 71,092 patients aged ≥60 years with type 2 DM and evaluated the relationships between baseline HbA1c levels and outcomes (acute nonfatal metabolic, microvascular, cardiovascular events, and mortality). The risk of any end point (complication or death) became significantly higher at HbA1c ≥8.0 % (64 mmol/mol). Patterns were generally consistent across age groups (60–69, 70–79, and ≥80 years). Our study focused on the care of poorly controlled elderly patients. Therefore, HbA1c level serves best to monitor their DM. The reduction in HbA1c level (−0.83 %) reported here may be beneficial because it suggests that a certified diabetes-educator pharmacist can significantly contribute to improving HbA1c values.

Our patients strictly adhered to their medications during consultation. Because a case-management nurse in the P4P program followed up on patients' ambulatory visits with telephone contacts, patients' medical records indicated regular prescription refills and ambulatory visits.

High unidentified cognitive deficit rates in older adults suggest the importance of periodic screening cognitive dysfunction [8]. Patients with such dysfunction have difficulties in performing complex self-care tasks, including taking drugs properly, changing insulin doses, or appropriately maintaining timing and dietary content. The pharmacist evaluated cognitive conditions of enrolled

patients using MMSE and found that MMSE scores did not differ between groups (25.4 ± 4.1 vs. 25.3 ± 3.2, $P = 0.52$). In older patients with cognitive dysfunction, we recommend simplified regimens, caregivers' involvement, and careful hypoglycemia monitoring. A well-trained pharmacist could educate patients and their families to prevent or resolve drug-related problems.

DM is associated with high depression prevalence. Untreated depression can lead to difficulty with self-care and implementing healthier lifestyle. Patient-initiated behaviors with regard to DM treatment (e.g., exercise, diet, medication adherence) are difficult to maintain in diabetic patients with major depression [26]. The pharmacist in our study identified 8 patients with depression in the pharmaceutical care group and referred these patients to the psychiatrist for further evaluation. If antidepressant therapy was initiated, the pharmacist discussed the importance of taking antidepressants with patients and provided consultation for using antidepressants.

In a previous comparative study of healthcare expenses [21], the P4P program increased physician visit costs, additional physical exams, laboratory tests, and medications under the regular NHI fee-for-service payment system; however, after accounting for the decreased hospitalization expenses, the increase in total expenses for diabetes-related care was not significantly higher than the standard care group. The care team of the P4P program improved the quality of care and helped to achieve a beneficial long-term outcome without increasing expenses significantly. Therefore, we expect that our findings will persuade healthcare policymakers to evaluate the cost and benefit of including a pharmacist case manager in the diabetes care team of the P4P program.

We believe that this study's success can be attributed to three critical components of pharmacist intervention: pharmacists who were qualified DM educators used their expertise to help patients and their families by providing knowledge regarding DM and resolving drug-related problems; pharmacists discussed recommendations for evaluating medication directly with primary care physicians and other diabetes care team members, which we consider an essential component of successful management programs; we used standardized screening tools for these patients, including MMSE for cognitive condition and TGDS-S for depression, thereby facilitating pharmacists to provide appropriate pharmaceutical care services to individuals.

This study has some limitations. First, a single center was studied for 1 year. Second, the study's general validity may be limited if P4P-program patients are not involved because it was designed to determine whether a pharmacist involved in the diabetes care team of the P4P program provided more benefits. Finally, although pharmacists without specialized training in DM and eldercare may provide limited benefits to patients with DM, the participating pharmacist involved here was well trained in diabetes care programs and was a certified DM educator with experience in providing elderly patients with, e.g., medication management and pharmaceutical care consultation. However, numerous pharmacists are qualified DM educators and are involved in the P4P program. We conclude that this study's results can be extended to diabetes care teams in other hospitals or clinics in Taiwan. Therefore, we think it will be beneficial to conduct a multi-center, long-term pharmaceutical care clinical trial for elderly patients with type 2 DM.

Conclusion

A pharmacist intervention program can provide pharmaceutical care services to poorly-controlled, ambulatory elderly patients with DM and improve HbA1c levels with safer and more effective use of medications. Pharmaceutical care did not significantly increase the burden of medical expenses. Our results suggest that well-trained pharmacists involved in multidisciplinary diabetes care teams can more effectively and safely improve diabetes control of the elderly.

Acknowledgments The authors thank the patients who participated in the study and the physicians, nurses (Huei-Yu Jheng), dietitians, and staff (Ker-Cheng Lin, Su-Mi Chen) at the Nantou Hospital, Ministry of Health and Welfare, Taiwan (ROC).

Funding This study was supported by a Grant from the Department of Health, Executive Yuan, Taiwan (ROC) (100-MID-08).

Conflicts of interest No potential conflicts of interest relevant to this article were reported.

References

1. American Diabetes Association. Standards of medical care in diabetes—2014. Diabetes Care. 2014;37:S14–80.
2. Wild S, Roglic G, Green A, Sicree R, King H. Global prevalence of diabetes: estimates for the year 2000 and projections for 2030. Diabetes Care. 2004;27:1047–53.
3. Mathers CD, Loncar D. Projections of global mortality and burden of disease from 2002 to 2030. PLoS Med. 2006;3:e442.
4. Centers for Disease Control and Prevention. National diabetes fact sheet: national estimates and general information on diabetes and prediabetes in the United States. Atlanta: US Department of Health and Human Services, Centers for Disease Control and Prevention; 2011.
5. Chang CH, Shau WY, Jiang YD, Li HY, Chang TJ, et al. Type 2 diabetes prevalence and incidence among adults in Taiwan during 1999–2004: a national health insurance data set study. Diabet Med. 2010;27:636–43.
6. Jiang YD, Chang CH, Tai TY, Chen JF, Chuang LM. Incidence and prevalence rates of diabetes mellitus in Taiwan: analysis of

the 2000–2009 nationwide health insurance database. J Formos Med Assoc. 2012;111:599–604.
7. Brown AF, Mangione CM, Saliba D, Sarkisian CA. Guidelines for improving the care of the older person with diabetes mellitus. J Am Geriatr Soc. 2003;51:S265–80.
8. Kirkman MS, Briscoe VJ, Clark N, Florez H, Haas LB, Halter JB, et al. Diabetes in older adults. Diabetes Care. 2012;12:2650–64.
9. Wubben DP, Vivian EM. Effects of pharmacist outpatient interventions on adults with diabetes mellitus: a systematic review. Pharmacotherapy. 2008;28:421–36.
10. McLean DL, McAlister FA, Johnson JA, King KM, Makowsky MJ, Jones CA, et al. A randomized trial of the effect of community pharmacist and nurse care on improving blood pressure management in patients with diabetes mellitus: study of cardiovascular risk intervention by pharmacists-hypertension (SCRIP-HTN). Arch Intern Med. 2008;168:2355–61.
11. Stone RA, Rao RH, Sevick MA, Cheng C, Hough LJ, Macpherson DS, et al. Active care management supported by home telemonitoring in veterans with type 2 diabetes: the DiaTel randomized controlled trial. Diabetes Care. 2010;33:478–84.
12. Mehuys E, Van Bortel L, De Bolle L, Van Tongelen I, Annemans L, Remon JP, et al. Effectiveness of a community pharmacist intervention in diabetes care: a randomized controlled trial. J Clin Pharm Ther. 2011;36:602–13.
13. Leal S, Glover JJ, Herrier RN, Felix A. Improving quality of care in diabetes through a comprehensive pharmacist-based disease management program. Diabetes Care. 2004;27:2983–4.
14. Simpson SH, Majumdar SR, Tsuyuki RT, Lewanczuk RZ, Spooner R, Johnson JA. Effect of adding pharmacists to primary care teams on blood pressure control in patients with type 2 diabetes. Diabetes Care. 2011;34:20–6.
15. Neto PR, Marusic S, de Lyra Junior DP, Pilger D, Cruciol-Souza JM, Gaeti WP, et al. Effect of a 36-month pharmaceutical care program on the coronary heart disease risk in elderly diabetic and hypertensive patients. J Pharm Pharm Sci. 2011;14:249–63.
16. McCord AD. Clinical impact of a pharmacist-managed diabetes mellitus drug therapy management service. Pharmacotherapy. 2006;26:248–53.
17. Cioffi ST, Caron MF, Kalus JS, Hill P, Buckley TE. Glycosylated hemoglobin, cardiovascular, and renal outcomes in a pharmacist-managed clinic. Ann Pharmacother. 2004;38:771–5.
18. Stading J, Herrmann J, Walters R, Destache C, Chock A. Impact of pharmacist intervention on diabetes patients in an ambulatory setting. Diabetes Spectr. 2009;22:241–6.
19. Guo NW, Liu HC, Wong PF, Liao KK, Yan SH, Lin KP, et al. Chinese version and norms of the mini-mental state examination. J Rehabil Med. 1988;16:52–9.
20. Mui AC. Geriatric Depression Scale as a community screening instrument for elderly Chinese immigrants. Int Psychogeriatr. 1996;8:445–58.
21. Lee TT, Cheng SH, Chen CC, Lai MS. A pay-for-performance program for diabetes care in Taiwan: a preliminary assessment. Am J Manag Care. 2010;16:65–9.
22. Chen HS. The effects of diabetes share care program in southern Taiwan. J Chin Med Assoc. 2011;74:283–4.
23. Faul F, Erdfelder E, Lang AG, Buchner A. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods. 2007;39:175–91.
24. Stratton IM, Adler AI, Neil HA, Matthews DR, Manley SE, Cull CA, et al. Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): prospective observational study. BMJ. 2000;321:405–12.
25. Huang ES, Liu JY, Moffet HH, John PM, Karter AJ. Glycemic control, complications, and death in older diabetic patients: the diabetes and aging study. Diabetes Care. 2011;34:1329–36.
26. Lin EHB, Katon W, Von Korff M, Rutter C, Simon GE, Oliver M, et al. Relationship of depression and diabetes self-care, medication adherence, and preventive care. Diabetes Care. 2004;27:2154–60.